

Mudasir Mujtaba

AI Engineer and Full-Stack Software Developer

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Portfolio: portfolio-mudasir-mujtaba-s-projects.vercel.app

Professional Summary

- AI Engineer and Full-Stack Software Developer and Computer Science graduate experienced in building and deploying production-grade AI systems and scalable web applications. Skilled in LLMs, RAG, Agentic AI, NLP, and Computer Vision, with full-stack development across the MERN and PERN stacks. First author of a research paper submitted to IEEE ICET 2026 on Retrieval-Augmented legal AI.

Technical Skills

- Languages:** Python, JavaScript, TypeScript, Java, PHP, SQL
- AI and Machine Learning:** Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, LLMs, RAG, Agentic AI, Prompt Engineering
- AI Frameworks:** PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, LangGraph, Hugging Face
- Full-Stack:** React.js, Next.js, Node.js, Express.js, FastAPI, Django, Flask, REST APIs, HTML5, CSS3, Tailwind CSS
- Databases:** PostgreSQL, pgvector, MongoDB, MySQL
- DevOps and Tools:** Docker, Kubernetes, MLflow, Git, GitHub Actions, CI/CD, JWT Authentication

Education

BS Computer Science

2022 – 2026

Sukkur IBA University, Sukkur

- CGPA: 3.5 out of 4.0. Final Year Project: LexiBot, an AI-Powered Legal Assistant for the Pakistani Legal Domain.

Publications

- LexiBot: A Retrieval-Augmented Legal Assistant Unifying Professional Research and Client Guidance for the Pakistani Legal Domain**
 - Authors: **Mudasir Mujtaba** (First Author), Mohsin Raza, Zakria, Muhammad Saddam Khokhar.
 - 21st IEEE International Conference on Emerging Technologies (IEEE ICET 2026), GIK Institute, November 2026. Status: Submitted, under review.
 - Developed a hybrid Retrieval-Augmented Generation pipeline over 29,657 Pakistani legal documents and 534,567 vector chunks, improving Recall at 5 from 0.640 to 0.813, with precedent-based outcome prediction at 60.4 percent accuracy (Macro F1 0.609).

Work Experience

Freelance AI/ML Engineer and Full-Stack Developer – Fiverr (Remote)

May 2024 – Present

AI/ML and Full-Stack Engineer

- Delivered AI solutions involving LLMs, RAG pipelines, NLP, Computer Vision, and intelligent automation for international clients.
- Developed and deployed production-grade AI APIs and full-stack web applications using FastAPI, Node.js, and Docker.

AI Development and Automation Intern – Techrypt (Karachi, Pakistan)

Jun 2025 – Aug 2025

AI and Full-Stack Developer

- Developed AI chatbots and automation workflows using LLMs and n8n for business process optimization.
- Contributed to CRM development through backend and full-stack feature implementation.

Freelance NLP Annotation Lead – CrowdGen (Remote)

Aug 2023 – Oct 2023

NLP Annotation Lead

- Worked on large-scale NLP projects with multilingual datasets (Sindhi, Urdu, English) supporting AI model training and evaluation.
- Led and coordinated data annotation workflows with distributed global teams to meet quality benchmarks.

Projects

LexiBot – AI-Powered Legal Assistant

Sep 2025 – May 2026

RAG and LLM Engineer

- Engineered a hybrid RAG pipeline over 29,657 Pakistani legal documents for legal research, case summarization, and outcome prediction with citation-backed answers.
- Built a multi-dimensional evaluation covering retrieval, faithfulness, attribution, and precedent voting. Work submitted to IEEE ICET 2026.
- GitHub: github.com/mudasirmujtaba/Lexibot-Legal-Assistant-AI-Application

SehatSaathi – Agentic AI Health Companion

2026 – Present

Full-Stack and Agentic AI Engineer

- Built an agentic AI web application that explains lab reports and prescriptions in plain language across English, Urdu, and Roman Urdu using OCR (Tesseract, PyMuPDF) and Google Gemini.
- Implemented RAG grounding over a medical knowledge base (pgvector), multilingual conversational question answering with context memory, per-user history, and responsible-AI safeguards.
- Stack: Next.js, TypeScript, FastAPI, PostgreSQL, pgvector, LangChain, Google Gemini, JWT, Docker.
- GitHub: github.com/mudasirmujtaba/SehatSaathi-Agentic-AI-Health-Companion

FaceTrack – Facial Recognition Attendance System

2026 – Present

Computer Vision and Full-Stack Engineer

- Built a facial-recognition attendance system that matches live camera frames against enrolled face embeddings (InsightFace SCRFD and ArcFace, 512-d) in under two seconds, with MediaPipe-based active liveness detection to reject printed photos and screen replays.
- Designed three independent services with automatic absent-marking, privacy-first design (biometric consent, irreversible embeddings instead of raw images, full data-deletion flow, and an admin audit trail).
- Stack: React 18, Vite, Tailwind (kiosk and admin dashboards); Python, FastAPI, InsightFace, MediaPipe; MongoDB; Docker; GitHub Actions CI.

Bachat Bazar – Full-Stack MERN E-Commerce Platform

2025

Full-Stack MERN Developer

- Built an e-commerce platform with MongoDB, Express, React, and Node.js, including a storefront and a JWT-authenticated admin dashboard.
- Implemented CRUD product and category management, cart, guest checkout, order management, and an analytics dashboard. GitHub: github.com/mudasirmujtaba/Bachat-Bazar-Ecommerce-Mern-Stack-Project

Certifications

- Claude with Amazon Bedrock – Anthropic and AWS — May 2026
- Large Language Models (LLMs) Concepts – DataCamp — Jun 2026
- Deep Learning and Generative AI – LinkedIn Learning — Nov 2025
- Google Prompting Essentials – Coursera (Google) — Jun 2026
- MLOps Essentials: Model Development and Integration – LinkedIn Learning — Dec 2025
- Software Architecture: Patterns for Developers – LinkedIn Learning — Dec 2025
- Python for Data Science, AI and Development – Coursera (IBM) — Feb 2025
- Developing Front-End Apps with React – Coursera (IBM) — Apr 2025
- Developing Back-End Apps with Node.js and Express – Coursera (IBM) — Apr 2025
- Introduction to MongoDB – MongoDB University — May 2025

Achievements

- Awarded the **HEC Need-Based Scholarship (Fall 2025-26)** by the Institutional Scholarship Award Committee, covering full tuition and a merit-linked semester stipend.
- **First author** of a research paper submitted to the 21st IEEE International Conference on Emerging Technologies (ICET 2026), currently under review.
- Maintained a strong academic record with a **CGPA of 3.5 out of 4.0** in BS Computer Science.